



TurnOne: The Cost of Convention in Low-Rank Games



Lucas Brennan-Almaraz
Department of Computer Science, Stanford University
CS 234: Reinforcement Learning — Winter 2026

Problem

Expert Pokémon VGC strategies are maximally different from Nash equilibria (TV = 0.99), yet expert-vs-expert payoffs match Nash within 0.02. Why does convention work in a ~200-action zero-sum game, and can offline RL beat it?

Pokémon VGC is a competitive doubles format where each player builds a team of six Pokémon, brings four to battle, and selects two as leads. On turn 1, both players simultaneously choose actions for their two active Pokémon. Each Pokémon can pick one of four moves targeted at either opponent or its partner, and the player may terastallize one Pokémon. This produces a combinatorial action space of ~200 valid combinations per side.

We formalize turn-1 play as an offline contextual bandit: context is the full game state, action is one of ~200 valid turn-1 combinations, and reward comes from a learned dynamics model. The game is zero-sum with payoffs derived from HP differences at KO and field state.

Data

We use 154,718 rated battles from Pokémon Showdown (Gen 9 VGC Regulation G, 1500+ Elo), yielding 309,436 directed turn-1 examples. Each example encodes full team composition (6 Pokémon with moves, abilities, items), lead assignments, and field state.

Approach

We compare three strategy types on the same reward model:

Behavioral Cloning (BC) imitates expert play and ignores reward. A Transformer encoder (4 layers, 4 heads, d=128) operates on 12 Pokémon tokens plus 1 field token. Two heads decode independently: a 16-way action head per active Pokémon and a 3-way tera head. The model achieves 44.8% top-1 accuracy (3× uniform) and 79.6% top-3.

Dynamics Model shares the encoder architecture, adds action embeddings fused with the encoder output, and predicts per-Pokémon HP changes, KO probabilities, and post-turn field state. This defines the bandit's reward function $r(s, a, a')$. It achieves HP MAE of 13.3/100 and KO AUC of 0.91.

Nash Equilibria are computed via LP on the dynamics model's payoff matrix $R \in \mathbb{R}^{m \times n}$. For each matchup, we enumerate valid actions per side and evaluate each pair with the dynamics model.

Conservative Q-Learning (CQL) is our offline RL baseline. It uses a factored Q-network $Q(s,a) = Q_a + Q_b + Q_{tera}$ with the same encoder, and penalizes Q-values on out-of-distribution actions. Because our horizon is one step, CQL reduces to conservative reward regression with no bootstrapping.

The puzzle: exploitable experts, zero payoff gap

Experts are exploitable by 1.41 reward units (~half a KO per turn) and play nothing like Nash (TV = 0.99). Nash puts mass on ~2.7 actions; BC spreads across ~95.

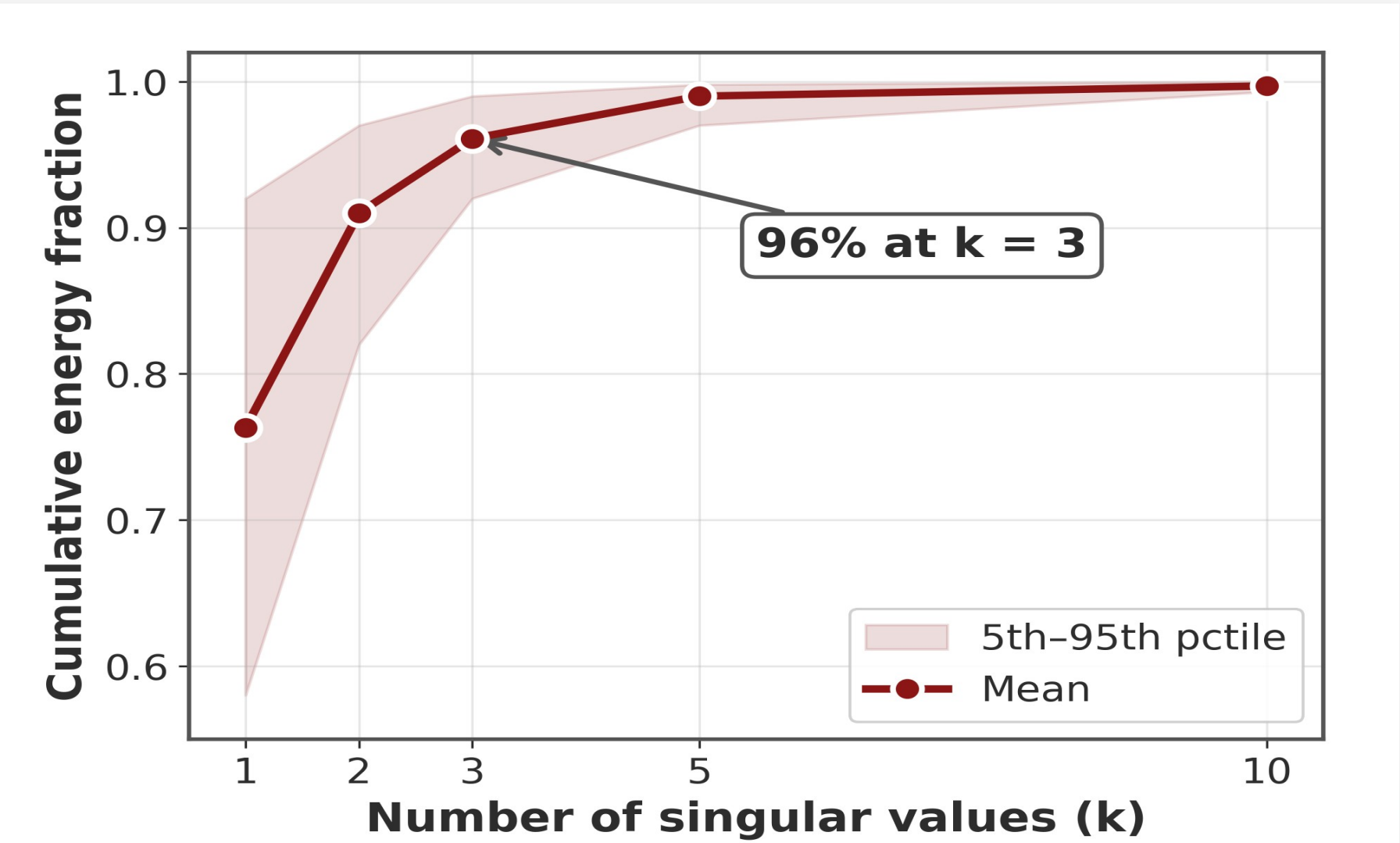
But expert-vs-expert payoff gap is -0.02, indistinguishable from zero. Convention is safe against itself and vulnerable to any optimizer.

Payoff landscape (500 matchups)	
Scenario	Payoff (P1)
BC worst-case	-1.19
Nash value V^*	+0.22
BC-vs-BC	+0.20
Best response to BC	+1.63
Exploitability	1.41
Cross-play gap	-0.02

Why convention is free: the game is 3-dimensional

The payoff matrix has effective rank 3 out of ~122 actions. Top 3 singular values capture 96% of spectral energy. Most of the ~200 nominal actions are functional duplicates that differ in name but produce similar outcomes.

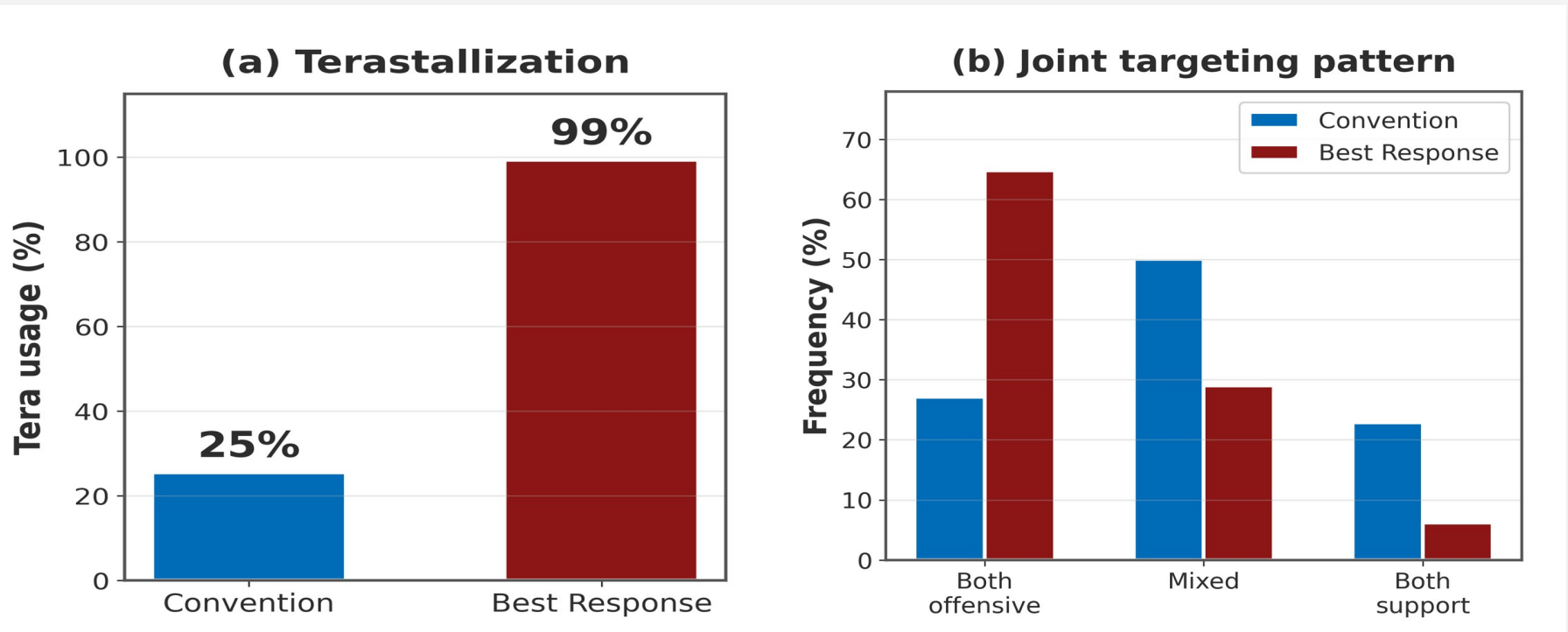
- 93% of BC-Nash deviation energy sits in the payoff null space
- Three effective dimensions: type advantage, raw damage, speed/priority
- Interaction decomposition: $\Delta_1 = +1.03$, $\Delta_2 = -1.02$, cross-term = -0.03. Deviations are orthogonal to R's column/row spaces, so they cancel.



Where convention fails: terastallization

Tera is irreversible: use it once and it's gone for the game. Experts conserve it for later turns (25% usage on turn 1), but myopic best response says use it now (99%).

- Tera alone accounts for 30% of total exploitability (0.41 of 1.41 reward units)
- Best responses also shift targeting: 65% both-offensive vs. 27% for BC



CQL: reward optimization can't escape convention

Switching from BC to CQL reshapes 60% of the action distribution but reduces exploitability by only 17%. Of the CQL-BC distributional shift, 94% lands in the payoff null space.

Metric	BC	CQL	Nash
Exploitability	1.40	1.15	0.00
TV(·, Nash)	0.99	0.98	0.00
Worst-case value	-1.23	-0.99	+0.16

- CQL doubles tera rate: 28% → 54% (Nash: 95%). Right direction, but the conservative penalty holds it back.
- CQL learns a different convention, not a better strategy.

Conclusions

- Payoff matrices have effective rank 3. 93% of strategic variation is payoff-irrelevant, because most actions are functional duplicates.
- The exception: terastallization, an irreversible resource, accounts for 30% of exploitability. Irreversibility is what makes convention costly.
- CQL confirms the bottleneck is game structure, not the learning objective. Reward optimization reshuffles the null space without escaping it.
- Prediction: convention should be free in games with low effective rank (RTS, sports playcalling) and costly where irreversible commitments dominate (poker all-ins, auctions).